genome to persist. This is challenging because the relative quiescence of stem cells makes them less susceptible to the molecular events of integration of a transgene or editing of a target gene. Yet, many of these technical challenges have been overcome sufficiently that there are medicines composed of hematopoietic stem cells bearing a gene to offset a missing or defective gene for active cerebral adrenoleukodystrophy, β thalassemia, and sickle cell anemia. These are termed gene therapies as a full coding sequence of the abnormal gene is provided. There is also a gene-edited therapy for β thalassemia and sickle cell anemia based on CRISPR-based technology. CRISPR-Cas allows for selected deletion of regions of genes and is thus called gene editing. In the case of the hemoglobinopathies, a gene that suppresses fetal hemoglobin, *BLC11A*, is the target. Reducing *BLC11A* expression by editing leads to fetal hemoglobin expression that overcomes the abnormal adult β hemoglobins causing β thalassemia and sickle cell anemia.

These therapies are powerful new ways in which hematopoietic stem cells can be medically beneficial. By modifying the patient's own cells, this type of hematopoietic stem cell therapy does not require immunologic matching with a donor and avoids graft-versus-host disease. At present, it still involves myeloablative conditioning of the patient prior to receiving the cells as they are modified outside the body and must compete with endogenous stem cells in the bone marrow. Efforts to enable gene editing in vivo and thereby avoid transplantation are being actively investigated.

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102 Iron Deficiency and Other Acquired Anemias Due to Decreased Erythrocyte Production

Clara Camaschella, Domenico Girelli

This chapter provides basic information on iron metabolism as an introduction to common anemias due to iron deficiency, inflammation, and aging, conditions in which alterations of iron balance play a prominent role. The pathogenetic mechanism of the anemia is hypoproliferation of the marrow precursors.

IRON METABOLISM

Iron is indispensable for multiple cell functions, including oxygen delivery, DNA replication and repair, metabolism, and energy production: hence, it is conserved and reutilized. However, to prevent potential toxicity due to the propensity of free iron to generate reactive oxygen species, iron is always bound to proteins, and its homeostasis is tightly regulated. Body iron cycle and distribution are illustrated in **Fig. 102-1**. Iron is provided to all cells by transferrin, a bilobed glycoprotein of hepatic origin abundant in plasma (204–360 mg/dL), that binds iron within a single (monoferric) or both lobes (diferric transferrin). Transferrin carries only 3–4 mg of iron but has a fast turnover. To produce about 200 billion erythrocytes daily, erythroid precursors need about 20–25 mg of iron, mainly supplied by macrophage recycling since dietary iron absorption is limited to 1–2 mg daily. Diferric transferrin binds with high affinity the iron importer transferrin receptor in all cells. The ligand-receptor complex is internalized via clathrin-coated pits to form the endosome. Released by low pH, iron is exported to cytosol by divalent metal transporter 1 (DMT1) and mainly used for heme and iron-sulfur cluster biosynthesis in mitochondria. Apotransferrin and transferrin receptor are recycled to cell surface and reutilized. Excess iron is safely sequestered by cytosolic ferritin, a large hollow shell formed by 24 (heavy and light) subunits in variable combination that stores iron into its central cavity, a process especially efficient in spleen macrophages. Ferritin is also secreted in blood where its concentration may be a marker of stored iron. In iron deficiency, cells can recover iron from ferritin by an autophagic process called ferritinophagy. Diferric transferrin also binds with lower affinity the iron sensor transferrin receptor 2 on hepatocyte **(Fig. 102-2*A*)** and erythroid cell surface. Erythroblasts express the highest number of transferrin receptors (~300,000–400,000/cell) and take up iron only from transferrin. Receptors not bound to transferrin are protease cleaved and released into blood where their concentration as soluble receptor is an estimate of total or iron-deficient erythropoiesis. Transferrin receptor is indispensable to all hematopoietic cells from stem cells to neutrophils. Lymphocyte function is strongly iron-dependent, as shown by the immunodeficiency resulting from rare mutations in transferrin receptor that hamper iron uptake. In the liver, transferrin receptor has the extra function of hepcidin regulator, binding and sequestering the hemochromatosis protein HFE (Fig. 102-2*A*). Liver, pancreas, and heart, but not erythroblasts, can take up non-transferrin-bound-iron (NTBI), which appears in the circulation when transferrin saturation exceeds 60%. No regulated mechanism for iron excretion exists. Physiologic iron loss occurs only through cell desquamation and menses in females and roughly balances the amount of daily absorbed iron (Fig. 102-1).

REGULATION OF IRON HOMEOSTASIS

Systemic iron homeostasis is regulated by *hepcidin*, a 25-amino-acid peptide mainly produced by hepatocytes that, blocking iron export and degrading the iron exporter ferroportin, inhibits plasma iron flux from duodenal enterocytes, iron-recycling macrophages, and iron-storing hepatocytes (Fig. 102-1). High hepcidin levels decrease, while low hepcidin levels increase, iron delivery to plasma. Hepcidin expression is increased by high plasma and liver iron that activate the BMP (bone morphogenetic protein)-SMAD (suppressors of mothers against decapentaplegic) signaling pathway. The proteins HFE, transferrin receptor 2, and hemojuvelin, whose mutations lead to different forms of hemochromatosis **(Chap. 426)**, contribute to hepcidin upregulation. Hepcidin is increased also by inflammatory cytokines, through interleukin (IL) 6 signal transducer and activator of transcription 3 (STAT3) pathway **(Fig. 102-2*A*)**.

Hepcidin expression is suppressed in iron deficiency by the hepatic transmembrane serine protease TMPRSS6 that antagonizes the BMP pathway. Another powerful inhibitor is erythroferrone **(Fig. 102-2*B*)**, the hormone secreted by erythroblasts upon erythropoietin stimulation, such as after a hemorrhage or in hypoxia, which sequesters the BMPs. Other hepcidin inhibitors include testosterone, which adapts iron supply to male erythropoiesis; fibrinogen-like protein 1 (FGL1), released by hepatocytes; and platelet-derived growth factor BB produced by many cell types in hypoxia.

At the cellular level, the iron regulatory proteins (IRPs) control the synthesis of transferrin receptor, ferritin, and ferroportin by interacting with the iron-responsive elements in their mRNA, thus coordinating cell iron uptake, with storage and efflux, according to the cell iron need. IRPs also tune erythropoiesis by regulating hypoxia-inducible factor (HIF) 2α and thus erythropoietin and the first enzyme of heme biosynthesis 5′-aminolevulinate synthase 2.

Intestinal Iron Absorption Iron needs change during life according to age and gender **(Table 102-1)**. In adults, only about 1–2 mg of iron are absorbed from the diet. Dietary iron content is closely related to caloric intake, with 6–7 mg elemental iron every 1000 calories: hence Western diet contains about 15 mg iron daily. The food composition

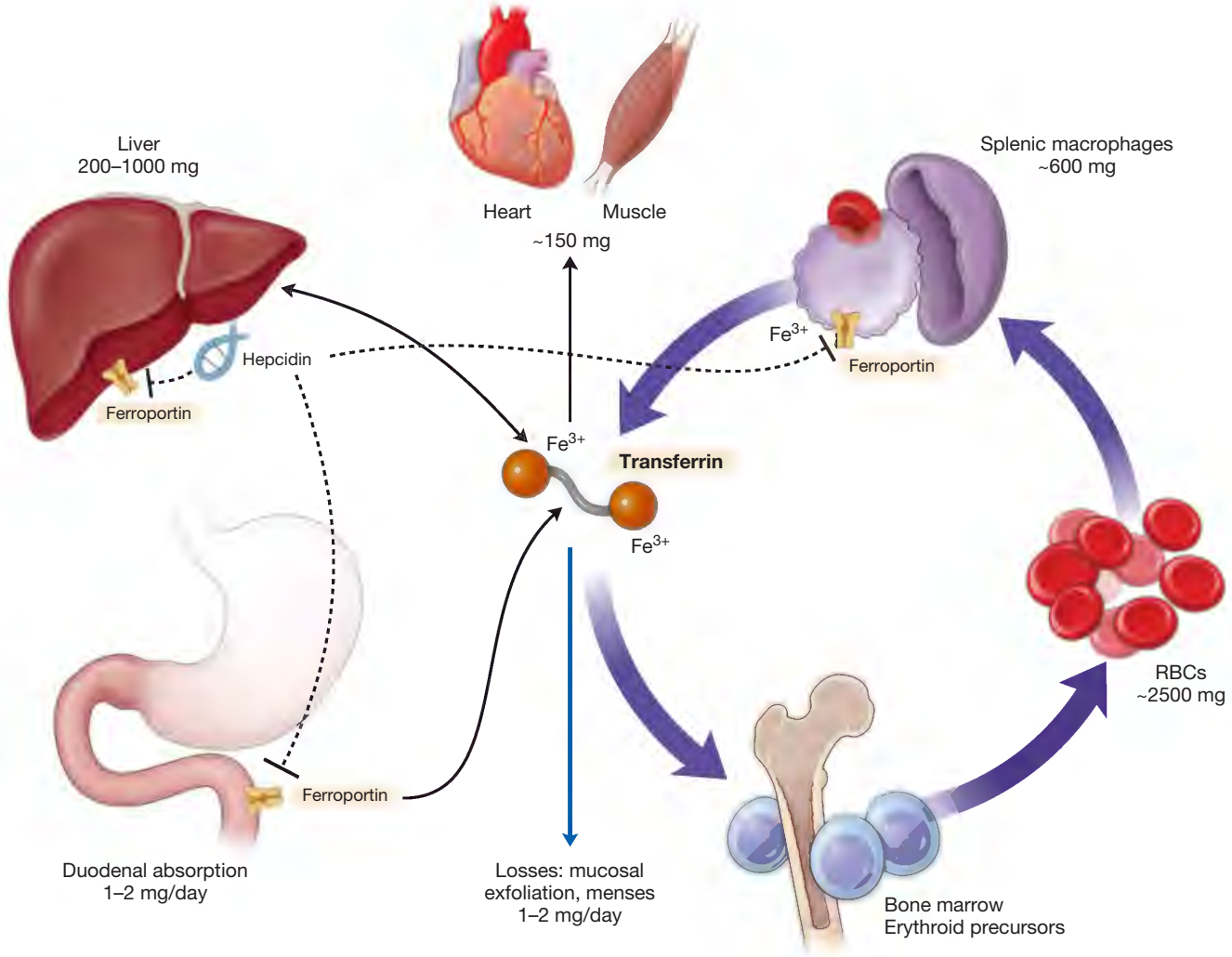

FIGURE 102-1 The iron cycle. Transferrin is central to iron trafficking. Most iron is recycled from macrophages, while only 1–2 mg are derived from intestinal absorption; mg indicates iron content of organs/tissues. Iron flux to plasma is controlled by hepcidin, a peptide produced by the liver that inhibits the activity of the iron exporter ferroportin in enterocytes, macrophages, and hepatocytes. Iron loss balances iron absorption.

affects iron bioavailability. Heme contained in red meat is easily taken up by enterocytes through a still undefined mechanism. Nonheme ferric (Fe^{3+}) iron is converted to ferrous (Fe^{2+}) iron by the low local pH and by the ferrireductase duodenal cytochrome B (DCYTB) before being imported from the lumen by DMT1. Phytates and phosphates contained in vegetables and cereals and tannin contained in tea and coffee reduce, while ascorbic acid enhances, inorganic iron absorption. Absorption takes place prevalently in duodenal enterocytes where iron can be utilized or released to plasma transferrin by ferroportin upon oxidation by hephaestin. Surplus iron is stored in cytosolic ferritin and lost by enterocytes sloughed into the lumen. Duodenal iron absorption depends both on the local levels of HIF-2α that enhance the expression of the absorption machinery components (DMT1, DCYTB, ferroportin) and on the hepcidin-mediated suppression of iron export. In iron deficiency, high levels of HIF-2α that favor luminal iron uptake and low hepcidin that facilitates export to plasma cooperate to maximize the absorption process. Iron in the gut may reshape the local microbiome, whose influence is emerging as modulator of body iron.

Macrophage Iron Recycling Specialized liver and spleen macrophages remove senescent erythrocytes (0.8–1% daily) and recycle to plasma the iron recovered from heme (Fig. 102-1). The high (20–25 mg daily) macrophage flux maintains steady-state iron supply to erythropoiesis and may further increase to compensate for hypoferremia or accelerated erythropoiesis. In contrast, decreased iron recycling, as caused by high hepcidin levels in inflammation, leads to anemia, with iron stored in macrophages. Because of its essential role in iron recycling, ferroportin is regulated at multiple levels, the most important being hepcidin-dependent. In intravascular hemolysis, hepatocytes and macrophages recover iron from hemoglobin-haptoglobin and heme-hemopexin complexes.

Iron Utilization Erythropoiesis utilizes a large amount of iron for heme/hemoglobin production and facilitates its own iron supply by releasing erythroferrone (Fig. 102-2*B*) upon erythropoietin stimulation. The release is transient; chronic erythroferrone production in ineffective erythropoiesis, as in non-transfusion-dependent β thalassemia (Chap. 103), leads to iron overload. In every cell, iron is mainly utilized in mitochondria to produce heme and iron/sulfur clusters, components of enzymes and complexes involved in the cell cycle, oxidative phosphorylation and energy production, fundamental processes in heart, skeletal muscle, and brain. Although bone marrow consumes most available iron, excessive erythroid use at the expense of other organs is prevented; one impeding mechanism is the iron export by erythroid ferroportin that contributes to serum iron levels in hypoferremia.

IRON DEFICIENCY AND IRON-DEFICIENCY ANEMIA

Iron deficiency and iron-deficiency anemia are global health problems affecting billions of people and a main cause of years lived with disability, especially among women. In isolated iron deficiency, iron stores are depleted while hemoglobin levels are still normal. In iron-deficiency

anemia, stores are depleted and iron is insufficient for erythropoiesis. Iron deficiency is *absolute* when the stores are exhausted and *functional* when erythropoiesis is iron-restricted but the stores are adequate or even increased.

PATHOPHYSIOLOGY

Isolated iron deficiency with low/absent iron stores is a common condition induced by rapid growth, inadequate nutrition, abundant menses. When iron is still sufficient for erythropoiesis, erythrocytes are normochromic and normocytic while ferritin is decreased (<15–30 μg/L). Isolated iron deficiency may progress to anemia of variable degrees. With transferrin saturation <15–20% iron supply to erythropoiesis is insufficient and erythrocytes become hypochromic and microcytic, the classical feature of iron-deficiency anemia. Though anemia is the most evident outcome, other organs, such as heart, muscles, and brain, are affected by iron deficiency. In chronic heart failure (CHF), iron deficiency causes symptoms even independently from anemia. Immune response may be blunted because lymphocytes require iron during the metabolic burst that mounts adaptative defense against infectious agents.

The relevance of low hepcidin in iron deficiency is clearly documented. Mutations of the negative regulator of hepcidin *TMPRSS6* cause a rare form of hereditary iron-refractory iron-deficiency anemia (IRIDA), characterized by reduced iron absorption and refractoriness to oral iron supplementation. Similarly, in inflammation, high hepcidin levels sequester iron into stores, leading to functional iron deficiency and iron-restricted erythropoiesis with low serum iron and transferrin saturation in the presence of elevated ferritin levels. Not infrequently, absolute and functional iron deficiency coexist, as in inflammatory bowel diseases, chronic kidney disease (CKD), and aging. Functional iron deficiency may be caused also by erythrocyte-stimulating agents (ESA) that, expanding erythropoiesis, unbalance iron demand and supply.

CAUSES OF IRON DEFICIENCY

Causes of absolute iron deficiency are detailed in Table 102-2. In general, iron supply is inadequate for iron needs because of excessive demand, insufficient intake, defective absorption, or blood loss. In clinical practice, multiple causes frequently co-occur, especially in the elderly and in patients with multimorbidity. Several genome-wide association studies showed that variants of iron genes, such as *TMPRSS6* and *TF* (encoding transferrin), may confer a genetic susceptibility to iron deficiency.

EPIDEMIOLOGY

Iron-deficiency anemia is estimated to affect ~1.2 billion individuals, roughly half of the world anemic population. The high iron requirements explain its high prevalence in preschool children (estimated ~25%), young women (37–40%), and pregnant women (>40% presenting with severe forms in low-income countries). Isolated iron deficiency is even more common, though precise figures are lacking. Both conditions are especially frequent in low-income countries, although high-income countries are not spared. In the United States, the prevalence of iron deficiency and iron-deficiency anemia in females aged 12–21 years based on ferritin <30 μg/L has been recently reported to be 38.6 and 6.3%, respectively. Anemia of inflammation,

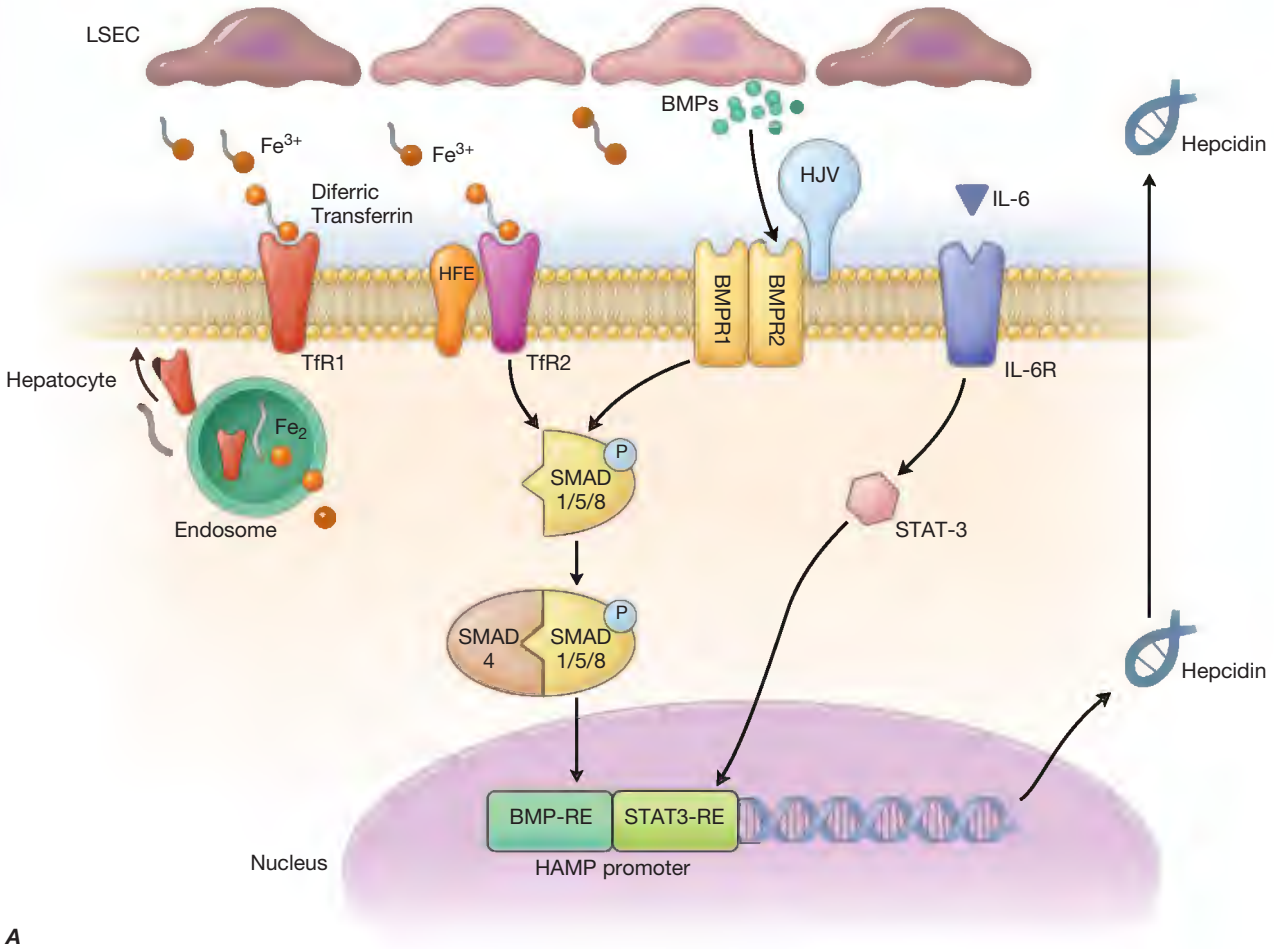

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FIGURE 102-2 A. Molecular regulation of hepcidin in iron deficiency. In steady state, diferric transferrin binds TFR1 to release iron to hepatocytes and interacts with TFR2 to signal iron concentration. Liver sinusoidal endothelial cells (LSECs) produce bone morphogenic proteins (BMPs) that interact with BMP receptors type 1 and 2 (BMPR1 and BMPR2, respectively) to activate cytosolic suppressors of mothers against decapentaplegic (SMAD) 1/5/8, with the cooperation of the coreceptor hemojuvelin (HJV) and HFE bound to TFR2. SMAD 1/5/8 complex with SMAD 4 and translocate to the nucleus, where they interact with BMP responsive elements (BMPRE) in hepcidin (*HAMP*) promoter to upregulate its expression. In inflammation, hepcidin is upregulated also through the interleukin 6 (IL-6)–IL-6 receptor–STAT3 pathway.

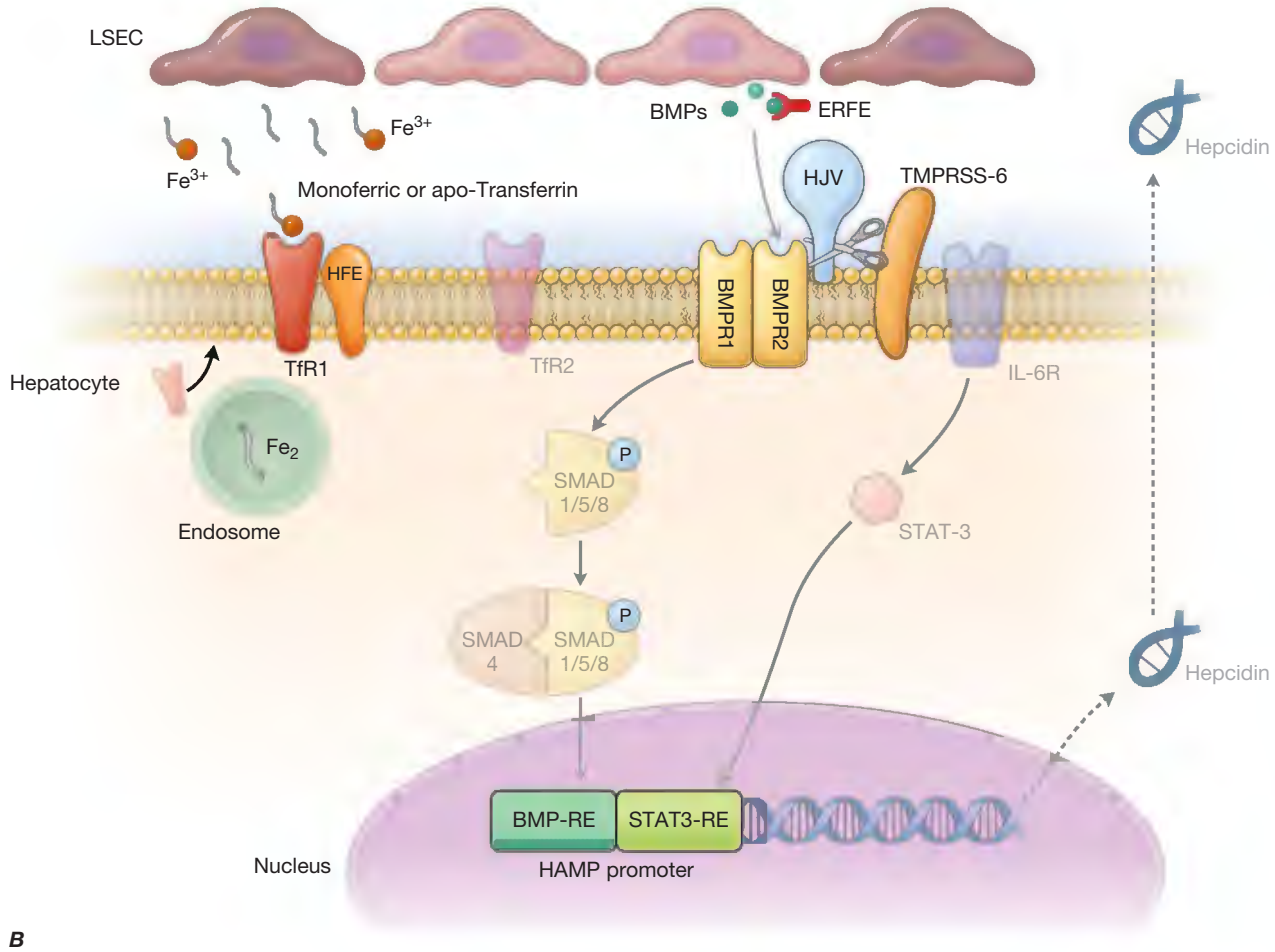

FIGURE 102-2 B. Molecular regulation of hepcidin in iron deficiency. In iron deficiency, transferrin is prevalently mono- or apotransferrin, the production of BMP by LSEC is scarce, the BMP coreceptor HJV is cleaved from the cell surface by the serine protease TMPRSS6, and TFR1 cannot transport iron inside the cell since it is bound to HFE. The SMAD and STAT3 pathways are inactive (indicated by *faint arrows*), and hepcidin is low/ undetectable.

due to functional iron deficiency, is the second most common anemia worldwide, especially in hospitalized patients, although precise figures are not available.

CLINICAL PRESENTATION

Anemia develops only in the late stage of iron deficiency, since the erythron is relatively spared compared to heart and skeletal muscle that may become iron depleted earlier. Because of slow progression and efficient compensatory mechanisms, anemia often remains undetected, especially in young people, until hemoglobin level falls below 8 g/dL. Isolated iron deficiency should not be underestimated since the resulting suboptimal metabolic function of myocytes, cardiomyocytes, and neurons may lead to fatigue, irritability, poor concentration, restless legs syndrome, and reduced exercise capability. As iron deficiency progresses, classical symptoms of anemia may appear (Chap. 66). Clinical clues pointing to iron-deficiency anemia include pica (craving and compulsive ingestion of nonnutritive substances such as ice, soil, or clay), nail fragility and koilonychia, hair loss, cheilosis, atrophic glossitis, and dysphagia caused by formation of esophageal webs (Plummer-Vinson syndrome). Clinical history is important to identify such symptoms as well as symptoms of any underlying disorder (Table 102-2).

TABLE 102-1 Daily Iron Requirements (mg) According to Age and Gender

Children	0.5–1.0
Adolescents	1.2–1.5
Adult males	1.0
Females (fertile age)	1.20–1.50
Pregnancy second trimester	2.8–3.0
Pregnancy third trimester	4.0–5.0
Breastfeeding	1.0-1.15
Females (postmenopausal)	1.0

Laboratory Iron Studies The diagnosis of iron deficiency with or without anemia is a two-step process that first requires laboratory tests, and then the identification of the cause(s) (Table 102-2). The laboratory diagnosis is usually simple, while tracing the cause may require imaging and endoscopy, procedures essential in males and postmenopausal females to rule out malignancy or other severe causes of bleeding. The absence of stainable iron in the bone marrow remains the gold standard for iron deficiency, although it is rarely needed and unacceptably invasive. The mainstay of absolute iron deficiency is low serum ferritin, which reflects reduced/exhausted iron stores. Although the precise cutoff is debated and varies in different settings, ferritin levels <30 μg/L have sufficiently high sensitivity and specificity (92 and 98%, respectively) to be used as a single test for diagnosing iron deficiency. Serum iron is reduced, but diurnal fluctuations prevent its use as a single test. Transferrin saturation (serum iron/total iron-binding capacity ratio) measures iron available for erythropoiesis, which becomes iron restricted for values <15–20%. Mean corpuscular volume (MCV) is <80 fL in established iron deficiency, but can be normal in early stages. Red cell distribution width (RDW) is often elevated, a measure of anisocytosis. Laboratory diagnosis may be demanding in patients with comorbidities. Classical microcytosis and hypochromia can be absent in elderly patients or masked by concomitant folate

TABLE 102-2 Causes of Absolute Iron Deficiency With or Without Anemia

MECHANISM	CONDITIONS
Increased iron needs	• Children, adolescents (growth spurt) • Pregnancy and breastfeeding • Treatment with erythropoiesis-stimulating agents
Inadequate iron intake	• Vegan or vegetarian diet • Malnutrition, poverty • Prolonged breastfeeding
Decreased iron absorption	• Gastrectomy, bariatric surgery • Celiac disease • *Helicobacter pylori* gastritis • Atrophic/autoimmune gastritis • Proton pump inhibitors[a] • Excess consumption of iron absorption inhibitors (e.g., tea, calcium) • IRIDA (*TMPRSS6* mutations)
Chronic blood losses	*Gastrointestinal* • Erosive esophagitis/gastritis • Peptic ulcer • Large hiatal hernia • Antral gastric ectasia (watermelon stomach) • Benign lesions (e.g., large polyps) • Meckel's diverticulum • Angiodysplasias (including Heyde's syndrome with concurrent aortic stenosis and acquired von Willebrand disease) • Malignancy (especially right colon cancer) • Large hemorrhoids • Inflammatory bowel diseases • Parasites (hookworms, others) *Gynecologic* • Heavy menstrual bleeding[b] • Fibroids • Uterine cancers *Urinary* • Intravascular hemolysis (e.g., PNH, prosthetic heart valves, march hemoglobinuria) • Urinary schistosomiasis (*Schistosoma haematobium*) • Bladder or renal cancer *Pulmonary* • Pulmonary hemosiderosis • Severe hemoptysis *Iatrogenic* • Hemodialysis • Surgical blood losses • Excess blood drawn for diagnostic testing • Drugs inducing or facilitating GI bleeding (aspirin, nonsteroidal anti-inflammatory drugs, corticosteroids, antithrombotic drugs) *Bleeding disorders* • von Willebrand disease, other coagulation defects, platelet disorders • Hereditary hemorrhagic telangiectasia (Rendu-Osler-Weber disease) *Frequent blood donors* *Self-inflicted in psychiatric disorders* (Munchausen's syndrome)

[a]Unlikely the sole cause. [b]May be exacerbated by undiagnosed von Willebrand disease or platelet disorders.

Abbreviations: IRIDA, iron-refractory iron-deficiency anemia; PNH, paroxysmal nocturnal hemoglobinuria.

and/or vitamin B_{12} deficiency in malabsorption disorders. **Table 102-3** lists all potentially useful laboratory parameters. Levels of soluble transferrin receptor increase in iron deficiency, since the receptor is highly expressed and released by erythroid precursors. Reduced reticulocyte hemoglobin content and increased percentage of hypochromic red cells reflect insufficient iron availability for erythropoiesis a few days or 2–3 months before testing, respectively. Serum levels of hepcidin are low/undetectable in absolute iron deficiency, a mechanism that maximizes dietary iron absorption. They are elevated in inflammation and in the rare IRIDA due to mutations of *TMPRSS6*. Overall, the clinical use of laboratory biomarkers other than ferritin and transferrin saturation is currently hampered by limited standardization and/or accessibility. Table 102-3 provides the differential diagnosis of iron deficiency based on iron parameters.

Severe iron-deficiency anemia may be accompanied by thrombocytosis due to prevalent megakaryocyte commitment of the common erythroid-platelet progenitor in the presence of insufficient iron.

TREATMENT

Both anemic patients and those with symptomatic isolated iron deficiency should receive iron replacement therapy. This includes oral and intravenous preparations **(Table 102-4)**, with different pharmacokinetic properties. The choice of the administration route depends mainly on the severity of anemia and the individual probability of iron absorption. To be efficacious, oral compounds require the integrity of the first digestive tract and full suppression of hepcidin. Iron salts, as iron sulfate, containing 65–100 mg of elemental iron per tablet, are the first choice in mild to moderate anemia, being widely available and cheap. Traditionally they were given once or twice daily, preferably before meals (Table 102-4). Gastrointestinal side effects, such as abdominal discomfort, nausea, epigastric burning, metallic taste, diarrhea, or stypsis, are common (30–60% of cases) and due to the toxicity of unabsorbed iron on the intestinal mucosa as a result of its limited absorption rate. The unabsorbed fraction can induce changes in the gut microbiome from harmless lactobacilli to pathogenic enterobacteria, promoting infectious diarrhea in children from low- to middle-income countries.

Another potential drawback that may reduce patient adherence to oral therapy is the need for prolonged administration, usually at least 3 months, to ensure recovery of hemoglobin levels and replenishment of iron stores to prevent early recurrence. Studies tracing iron absorption with stable isotopes are changing the paradigm of oral iron administration. An alternate-day schedule maximizes iron absorption as compared to daily administration, which induces a transient peak of hepcidin; moreover, it is better tolerated. Addition of 80 mL of orange juice (but not of pharmacologic doses of ascorbic acid) maximizes the fraction of absorbed iron, indicating that the best regimen is alternate-day iron salts plus "natural" vitamin C. Lack of reticulocytosis after 1 week or a <1 g/dL hemoglobin increase after 2 weeks of treatment defines the response as inadequate and should prompt a search for the cause, such as lack of compliance, ongoing blood loss, or malabsorption.

Intravenous iron preparations (Table 102-4) are best used in case of intolerance to oral iron or malabsorption and for rapid correction of severe anemia (hemoglobin ≤8 g/dL). These nanodrugs share a common structure with a shell of carbohydrate moieties surrounding an iron hydroxide core. Their use has long been limited by safety concerns, especially anaphylaxis reactions that occurred with early preparations, such as the high-molecular-weight dextran, that are no longer on the market. The latest generation of compounds show a good safety profile, do not elicit IgE-mediated response, and are highly stable, allowing the administration of the therapeutic dose in a single infusion or two infusions. A dose test is no longer required. The nanoparticles are taken up by macrophages, and iron is slowly released to transferrin through ferroportin. Infusion reactions may occur if a small fraction of iron is rapidly released from the nanoparticle core, depending on the stability of the shell. Reactions occur in <0.1% of cases and are usually mild and self-limited. More severe reactions, possibly complement mediated (complement activation–related pseudo-allergy), are extremely rare. Other side effects are flu-like symptoms 2–5 days after the infusion and hypophosphatemia, more commonly reported with ferric carboxymaltose and mediated by the increase of the phosphaturic hormone fibroblast growth factor 23. Hypophosphatemia is

TABLE 102-3 Laboratory Biomarkers for Differential Diagnosis of Iron Deficiency

	IRON DEFICIENCY	ANEMIA OF INFLAMMATION	MIXED FORMS	THALASSEMIA TRAIT
Ferritin (μg/L)	Low (≤15–30)	Normal to high (>100)	Low to normal (<70–100)	Normal to high
Serum iron (μmol/L)	Low	Low	Low	Normal to high
Transferrin saturation (%)	Low (<10)	Low (<20)	Low (<20)	Normal to high
MCV (fL)	Low (<80)	Low to normal	Low to normal	Low
RDW	High	Normal	Normal to high	Low to normal
% HYPO	High (>6%)	High	High	High
CHr	Low	Low	Low	Low
sTfR	High	Low to normal	Normal to high	Normal to high
sTfR/log ferritin	>2	<1	>2	NA
Hepcidin[a]	Very low	Normal to high	Low	Low to normal
Zn-PP[b] (μM/mol heme)	High (>80)	Normal to high	High (>80)	High

[a]Serum hepcidin levels are not routinely measured; reference values are not standardized. [b]Zn-PP indicate zinc protoporphyrins, which reflect iron/zinc competition in the last step of heme synthesis; used in screening programs.

Note: % HYPO (percentage of hypochromic red blood cells) and CHr (hemoglobin content of reticulocytes) are measured by automated analyzers and available upon request.

Abbreviations: MCV, mean corpuscular volume; NA, not assessed; RDW, red cell distribution width; sTfR, soluble transferrin receptor.

transient and asymptomatic in most cases, whereas patients undergoing multiple high-dose infusions or with micronutrient deficiencies are at risk of developing osteomalacia and bone fractures.

Overall, the latest generation of compounds have revolutionized iron supplementation because of easy administration, reduction of hospital visits, and reassuring safety profile. Consequently, indications have been expanded to treat severe anemia in the second and third trimesters of pregnancy and in complex conditions, such as CHF, perioperative anemia, and anemia in the emergency department (see below).

Iron-deficiency anemia should not be treated with blood transfusions unless patients have hemodynamic instability or are rapidly deteriorating.

TABLE 102-4 Iron Preparations for Replacement Therapy

Oral Compounds

	ELEMENTAL IRON CONTENT,[a] mg	DOSAGE
Iron sulfate	65–105	65–100 mg 1–2 times a day or 65–100 mg on alternate day
Iron gluconate	27–38	1–2 doses per day
Iron fumarate	29–105	1–2 doses per day
Sucrosomial iron[b]	30	30 mg/d
Polysaccharide iron complexes (tablet or elixir)	50–200 15/mL 125/mL	50–200 mg/d
Ferric maltol	30	30 mg 2 times a day
Ferric citrate	45–210	3 doses per day in chronic kidney disease patients with hyperphosphatemia

Intravenous Compounds

	IRON CONTENT, mg/mL	MAXIMUM SINGLE DOSE
Iron sucrose	20	200–300
Iron gluconate	12.5	125, slow infusion
Low-molecular-weight dextran	50	1000, over 1 h
Iron carboxymaltose	50	750–1000, over 15 min infusion
Iron derisomaltose	100	Up to 20/kg, rapid infusion
Ferumoxytol	30	510, rapid infusion

[a]The content of *elemental* iron in sulfates, gluconate, and fumarate varies from 10 to 33% (e.g., an iron sulfate tablet contains 100 mg of *elemental* iron in 325 mg of the total salt).

[b]Sucrosomial iron consists of ferric pyrophosphate conveyed by a phospholipid and sucrester matrix.

PREVENTION

Prevention is fundamental in low- to middle-income countries where iron-deficiency anemia is common due to the combination of malnutrition and increased blood loss by helminth infections. In this setting, prevention of iron deficiency at the population level employing food fortification or oral iron administration must take into account the infectious risk. Iron deficiency confers some protection from malaria in endemic areas, whereas iron supplementation may favor malaria progression by enhancing parasite trophism for reticulocytes that are increased during anemia recovery. In view of such bidirectional association, in malaria-endemic regions, the World Health Organization recommends iron supplementation only when concomitant strategies to prevent and treat malaria are implemented. Proposed preventive strategies for pregnancy to improve maternal and infant outcome are either routine prenatal supplementation or regular checking of hemoglobin/iron status and on-demand treatment. Frequent blood donors are another population at risk, considering that each unit of blood contains ~200 mg of iron. Prolonging the interdonation interval and monitoring ferritin levels are prevention measures, especially in female donors, who may eventually require iron supplementation. Iron deficiency in patients undergoing major surgery predisposes to anemia due to blood loss during the procedure and the frequent diagnostic testing. Preoperative screening and treatment of iron deficiency is one pillar of patient blood management strategies that aim to reduce perioperative complications and length of hospital stay and save cost. Intravenous route is preferred if surgery is within 6 weeks from the diagnosis of iron deficiency.

ANEMIA OF INFLAMMATION

Anemia of chronic diseases indicates the multifactorial anemia observed in a heterogeneous group of disorders, including infections, autoimmunity, malignancies, and inflammatory bowel diseases, that share mechanisms of immune activation and cytokine production. Similar mechanisms account for the anemia that develops in acute conditions such as sepsis, severe trauma, and after major surgery so that the broader term *anemia of inflammation* is now preferred. This type of anemia spans the whole spectrum of clinical inflammatory disorders, from hyperacute, exemplified by macrophage activation syndrome, to subclinical, as in CHF, diabetes, and obesity.

PATHOPHYSIOLOGY

At least three cytokine-driven mechanisms account for the diminished red cell production in inflammation (Fig. 102-3). Increased hepcidin levels, mainly induced by IL-6, cause macrophage iron sequestration and restrict iron reutilization. Erythropoietin (EPO) production is suppressed by IL-1β and tumor necrosis factor α (TNF-α). Response of erythroid progenitors to EPO is blunted by interferon-γ (IFN-γ)

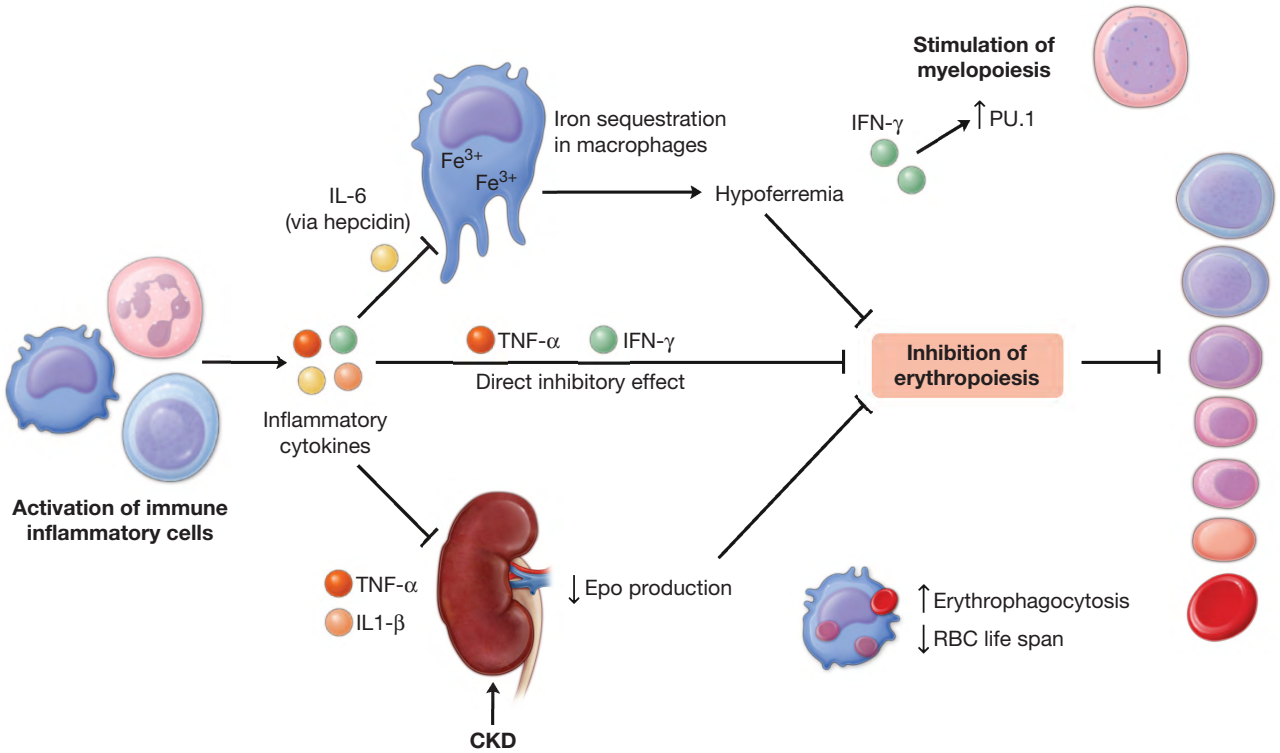

FIGURE 102-3 Pathophysiology of anemia of inflammation. Activation of immune cells leads to the production of inflammatory cytokines. IL-6 induces macrophage iron sequestration via hepcidin, tumor necrosis factor α (TNF-α) and interferon γ (IFN-γ) reduce erythropoietin (EPO) production and erythropoietic response; activation of PU.1 transcript induces myelopoiesis. Hemolysis of red cells phagocytosed by macrophages variably contribute to the pathophysiology of anemia, especially in acute inflammation.

and TNF-α. Moreover, bone marrow progenitor commitment is reprogrammed toward the myeloid lineage through increased activity of PU.1 and other transcription factors. In acute inflammation, immature white blood cells may be released to the circulation, the so-called "emergency myelopoiesis," as observed in severe COVID-19. A mild decrease of erythrocyte life span contributes to anemia. Iron sequestration in stores and hypoferremia are central to anemia of inflammation, emphasizing that it is a disorder of iron distribution. The contribution of the other mechanisms varies in different conditions. EPO deficiency is key in CKD. Massive red cell destruction is the main driver of anemia only in hyperinflammation, such as Still's disease and hemophagocytic lymphohistiocytosis. Mechanistically, hepcidin hyperexpression and hypoferremia are part of the innate immune response, aimed at depriving iron from and blocking the growth of the invading pathogens. However, in the long term and in noninfectious disorders, hypoferremia becomes a maladaptive response that leads to anemia.

CLINICAL FINDINGS AND DIAGNOSIS

Anemia of inflammation is mild to moderate, normocytic, and normochromic with low reticulocyte count. It can become microcytic when longstanding, due to iron-restricted erythropoiesis. It can be asymptomatic or contribute to disease burden, worsening the prognosis, as in CHF and CKD. A biochemical hallmark of inflammation is hyperferritinemia (Table 102-3). Ferritin, especially its heavy chain, is an acute phase reactant; thus, it cannot be used as an iron marker in inflammation. The combination of hypoferremia and hyperferritinemia in inflammatory disorders defines "functional" iron deficiency, indicating insufficient erythroid iron notwithstanding the presence of adequate stores. The boundaries between absolute and functional iron deficiency are not unequivocal, as shown by inflammatory bowel diseases, colon cancer, CKD, and CHF. Detecting true iron deficiency in these settings is challenging as both a universal definition and biomarkers are lacking. Ferritin <100 μg/L is the most accepted diagnostic threshold. In CHF and CKD, absolute iron deficiency is also defined by transferrin saturation <20%, irrespective of ferritin levels. A proposed diagnostic tool is the soluble transferrin receptor (sTfR)/log ferritin ratio (Table 102-3). The clinical usefulness of other laboratory parameters needs confirmation.

In CKD, anemia commonly develops when >50% of kidney function is lost and progresses in parallel with the reduction of glomerular filtration rate. It is a negative prognostic factor, predisposing to cardiovascular complications, hospitalization, and mortality. EPO deficiency, systemic inflammation, and negative iron balance are key contributors to the complex pathophysiology of anemia in CKD. Impaired iron absorption, chronic bleeding due to platelet dysfunction, and blood loss in the dialysis apparatus are causes of absolute iron deficiency, whereas increased production and decreased clearance of hepcidin result in functional iron deficiency. Deficiency of folate and vitamin B_{12} and a hemolytic component in uremic patients may contribute to anemia.

In CHF patients, the prevalence of anemia and iron deficiency is reported to be as high as 50–60% with negative prognostic impact. Besides mild inflammation, mechanisms include malnutrition, reduced iron absorption, and blood loss caused by the use of antithrombotic drugs. Current guidelines recommend periodic screening for iron deficiency in CHF patients. Supplementing intravenous iron may improve symptoms and quality of life, although influence on mortality is not proven.

Anemia is common in cancer, especially in hematologic malignancies such as myelodysplasia and multiple myeloma. It is multifactorial as it may be associated with iron and vitamin deficiency, blood loss, and chemotherapy side effects. Moderate to severe anemia is frequent in critically ill patients and is associated with unfavorable outcomes such as prolonged hospitalization and even mortality. Nearly 50% of patients admitted to an intensive care unit (ICU) have a hemoglobin level <9 g/dL, and most discharged survivors have persistent anemia. Anemia in ICU reflects anemia of inflammation, with the addition of blood losses due to blood sampling (up to 40 mL/d) for laboratory monitoring.

A mild anemia is common in elite endurance athletes, especially females. It is multifactorial and due to reduced iron absorption, iron losses, and increased hepcidin caused by both mild inflammation and indirect activation of the hepatic SMAD pathway by lactic acid.

TREATMENT

Treatment of the underlying disorder controls anemia of inflammation, as shown by the effect of anti-IL-6 agents in Castleman's disease and rheumatoid arthritis. Standard treatment of anemia in CKD is based on ESAs **(Chap. 322)** with a target hemoglobin established at <11.5 g/dL to avoid cardiovascular complications. Since iron deficiency leads to ESA hyporesponsiveness, iron supplementation is usually required, mostly parenterally, because in advanced CKD high hepcidin levels prevent iron absorption. Intravenous iron is administered at high or refracted doses, ideally to maintain transferrin saturation at 30–40% and ferritin between 200 and 700 μg/L. Iron supplementation may lower the dose of ESA required to maintain the hemoglobin target. An oral alternative to intravenous iron is ferric citrate, which, binding intestinal phosphate, provides iron while reducing phosphate levels. Prolyl hydroxylase inhibitors that stabilize HIF-2α have been approved in some countries with the aim of replacing with a single oral drug both ESA and iron, since HIF upregulates EPO production as well as duodenal iron absorption. Although these novel antianemic drugs are efficacious, safety concerns still exist for their long-term use.

Intravenous iron can be used in cancer-related anemia to reduce the need of transfusions, especially with ferritin <100 μg/L and before elective surgery. ESAs are used in low-risk myelodysplastic syndromes and in patients receiving chemotherapy notably when cancer is deemed uncurable. Intravenous iron supplementation is currently recommended in CHF patients with either ferritin <100 μg/L or transferrin saturation <20% with ferritin of 100–300 μg/L. Nonetheless, there is still some uncertainty as to whether this represents the best definition of iron deficiency in CHF. Novel drugs targeting the hepcidin-ferroportin axis to counteract iron maldistribution are under investigation.

Blood transfusions remain the therapeutic option for severe anemia in the ICU, with guidelines increasingly recommending restrictive hemoglobin thresholds (7–8 g/dL) for hemodynamically stable patients. Intravenous iron is proposed to minimize the use of transfusions, with some studies reporting improved clinical outcomes, although the risk of infections and hypophosphatemia are of concern in these fragile patients.

ANEMIA OF AGING

Prevalence of anemia in the elderly is ~10% at the age of 65 years, 25% in people >85 years, and 50% in those affected by multimorbidity, hospitalized, or institutionalized. Anemia is typically mild (hemoglobin 11–12 g/dL) and multifactorial, and its impact is frequently underestimated. Mounting evidence strengthens its negative influence on quality of life, muscle weakness, and the risk of falls and fractures, as well as its independent association with mortality. The adverse effects of anemia are seen in black patients at hemoglobin levels ~1 g/dL lower than in white patients.

Approximately one-third of the cases are estimated to be due to nutritional causes, especially iron deficiency; one-third are due to inflammation, including CKD; and one-third remain without a detectable origin and are defined "unexplained anemia of aging." Age-related decline in testosterone levels in males, stem cell exhaustion, and EPO reduction might contribute. Absolute iron deficiency is common, due to poor dietary iron intake, impaired absorption by chronic gastritis, or the use of proton pump inhibitors and blood losses. A gastrointestinal malignancy should always be ruled out in anemic elderly patients, but chronic bleeding may also result from nonmalignant intestinal lesions such as angiodysplasias and the use of antithrombotic drugs for common disorders such as atrial fibrillation. A recent study has shown that 100 mg of daily aspirin given for primary cardiovascular prevention in elderly subjects was associated with a 20% higher risk of anemia and iron deficiency as compared to placebo. The proposed ferritin cutoff for iron deficiency (45–70 μg/L) is higher than in younger patients. Treatment of iron deficiency may mitigate the negative prognostic implications of anemia.

Functional iron deficiency due to overt or subclinical inflammation often contributes, complicating the laboratory diagnosis. Traditionally "bone marrow senescence," now known as clonal hematopoiesis of indeterminate potential (CHIP), is present in >10% of otherwise-healthy individuals aged 70 years and increases sharply to >60% in the eighties. CHIP is associated with a chronic low-grade inflammatory status that worsens anemia and cardiovascular risk. Patients should be followed up for the risk of developing myelodysplastic syndromes **(Chap. 107)**. The unexplained anemia of aging is usually characterized by an EPO level that is lower than expected for the degree of anemia. EPO can increase the hemoglobin level, but it is unclear whether that change has an influence on the hospitalization, frailty, and mortality risks associated with the anemia.

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103 Disorders of Hemoglobin

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Hemoglobinopathies result from changes in the amino acid sequence of globin; in thalassemia, synthesis of normal globin is insufficient. Together the disorders of hemoglobin compose the most common Mendelian genetic diseases. In addition, they are responsible for most instances of hemolytic anemia. Sickle cell disease and the hemoglobin E (HbE)–associated syndromes are the most prevalent hemoglobinopathies; β and α thalassemia are the most prevalent thalassemias. In addition to these common disorders of hemoglobin, rare globin mutations cause hemoglobin instability, cause increased or decreased affinity of hemoglobin for oxygen (O_2), and allow spontaneous oxidation of hemoglobin, reducing its O_2 transport. O_2 transport can also be reduced by exposure to carbon monoxide (CO) and extrinsic oxidizing agents **(Table 103-1)**.

HEMOGLOBIN

Easy access to erythrocytes to study hemoglobin structure and function, reticulocytes to examine hemoglobin biosynthesis, and leukocyte DNA to define the mutations of hemoglobin, with the availability of hematopoietic stem and progenitor cells from blood and bone marrow, has placed hemoglobin disorders in the forefront of molecular medicine. The biology of hemoglobin provides the background for understanding the pathophysiology of its many genetic and acquired disorders and approaches to their treatment.